

# Disproof of Conjecture 00000002141: there are two different cospectral pairs of 4-regular graphs of order 10

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2026-10-04

## Abstract

Conjecture 00000002141 asserts that the minimal order of a pair of cospectral (non-isomorphic) regular graphs is 10, and that the cospectral regular pair of order 10 is unique. We exhibit two cospectral pairs  $(A_1, B_1)$  and  $(A_2, B_2)$  of 4-regular graphs on 10 vertices. The two pairs have different spectra, and no graph of the first pair is isomorphic to a graph of the second. So the uniqueness clause is false. An exhaustive enumeration shows that order 10 has exactly two 4-regular cospectral pairs and their two 5-regular complements. The disproof is formalized in Lean 4 (core library only).

## 1 The conjecture

*Definition:* Cospectral regular graphs are non-isomorphic regular graphs with the same adjacency spectrum. *Conjecture:* The minimal order of a cospectral regular pair is 10 (the Saltire pair); the cospectral regular pair of order 10 is unique (classification uniqueness).

The Chinese version says the same (the order-10 pair is the only one). The conjecture is a conjunction, so it suffices to refute the uniqueness clause. We read “the pair is unique” in the weakest sense: any two cospectral regular pairs  $\{G, H\}$  and  $\{G', H'\}$  of order 10 coincide up to isomorphism as unordered pairs. (The first clause is in fact true; see Section 5. The name “Saltire pair” usually refers to the non-regular pair  $K_{1,4}, C_4 \cup K_1$  on 5 vertices, but this plays no role.)

## 2 The graphs

All four graphs have vertex set  $\{0, \dots, 9\}$  and 20 edges.

$A_1$  : 03, 05, 07, 09, 14, 15, 17, 19, 25, 26, 27, 28, 36, 38, 39, 46, 48, 49, 57, 68

$B_1$  : 03, 05, 07, 08, 14, 15, 18, 19, 25, 26, 27, 29, 36, 37, 38, 46, 48, 49, 57, 69

$A_2$  : 03, 05, 07, 08, 14, 15, 17, 19, 25, 26, 28, 29, 36, 37, 38, 46, 48, 49, 57, 69

$B_2$  : 03, 04, 06, 07, 15, 16, 18, 19, 25, 27, 28, 29, 34, 36, 37, 48, 49, 57, 58, 69

(In graph6 format: ICR‘vGyu?, ICR‘tjWY\_, ICR‘uiwY\_, ICdedhkY\_.) Each vertex has degree 4 in each graph.

### 3 Cospectrality

For a graph with adjacency matrix  $A$  and eigenvalues  $\lambda_1, \dots, \lambda_n$ , the number of closed walks of length  $k$  is  $\text{tr}(A^k) = \sum_i \lambda_i^k = p_k$ . By Newton's identities  $ke_k = \sum_{i=1}^k (-1)^{i-1} e_{k-i} p_i$ , the power sums  $p_1, \dots, p_n$  determine the elementary symmetric functions  $e_1, \dots, e_n$ , hence the characteristic polynomial  $\prod (x - \lambda_i)$ , hence the spectrum. The converse is clear. So two graphs on  $n$  vertices are cospectral if and only if they have equally many closed walks of each length  $k \leq n$ .

**Proposition 1.**  $\text{tr}(A_1^k) = \text{tr}(B_1^k)$  and  $\text{tr}(A_2^k) = \text{tr}(B_2^k)$  for  $0 \leq k \leq 10$ . Equivalently, the characteristic polynomials agree; they are

$$\begin{aligned} \det(xI - A_1) &= \det(xI - B_1) = x^{10} - 20x^8 - 16x^7 + 110x^6 + 136x^5 \\ &\quad - 180x^4 - 320x^3 + 9x^2 + 200x + 80, \\ \det(xI - A_2) &= \det(xI - B_2) = x^{10} - 20x^8 - 14x^7 + 108x^6 + 104x^5 \\ &\quad - 183x^4 - 188x^3 + 80x^2 + 68x - 16. \end{aligned}$$

The two polynomials differ. For example,  $\text{tr}(A^3)$  is six times the number of triangles, and  $\text{tr}(A_1^3) = 48$  (8 triangles) while  $\text{tr}(A_2^3) = 42$  (7 triangles).

*Proof.* Direct computation, in Lean (`cospectral_1`, `cospectral_2`, `spectra_differ`) and in `verify.py`. The script computes the characteristic polynomials in two independent exact ways: Faddeev–LeVerrier, and  $\det(xI - A)$  by exact Gaussian elimination at  $x = 0, \dots, 10$ .  $\square$

**Certificates independent of Newton's identities.** Each pair is related by a Godsil–McKay switching [1] on the set  $D = \{0, 4, 5, 6\}$ , followed by a relabelling. Concretely, Lean checks integer matrices  $M_1, M_2$  with  $M_i M_i^\top = 4I$  and  $M_i A_i = B_i M_i$ . Thus  $Q_i = M_i/2$  is orthogonal and  $B_i = Q_i A_i Q_i^\top$ , so  $A_i$  and  $B_i$  are orthogonally similar and have the same spectrum (`orth_1`, `orth_2`). Lean also computes the coefficients of the characteristic polynomials displayed above from the closed-walk counts (`cp_A1`,  $\dots$ , `cp_B2`).

### 4 Non-isomorphism

Call an edge  $uv$  *triangle-free* if  $u$  and  $v$  have no common neighbour. Both of the following properties are preserved by graph isomorphisms:

- (P) some edge is triangle-free;
- (Q) some triangle-free edge shares no endpoint with any other triangle-free edge.

**Lemma 2.**  $A_1$  fails (P), while  $B_1$ ,  $A_2$  and  $B_2$  satisfy (P).  $A_2$  satisfies (Q), while  $B_2$  fails (Q).

*Proof.* The triangle-free edges are: none in  $A_1$ ; 15, 36 in  $B_1$ ; 25, 28, 36, 48 in  $A_2$ , where 36 is isolated; and 29, 48, 49 in  $B_2$ , which form the path 2–9–4–8.  $\square$

All four graphs are connected. This is forced anyway: the only disconnected 4-regular graph on 10 vertices is  $2K_5$ , which has eigenvalue 4 with multiplicity 2 and so has no cospectral mate. Hence  $A_1 \not\cong B_1$  and  $A_2 \not\cong B_2$ , so both are cospectral regular pairs. Also  $A_1 \not\cong A_2$  and  $A_1 \not\cong B_2$ , so the unordered pairs  $\{A_1, B_1\}$  and  $\{A_2, B_2\}$  are different up to isomorphism. The two pairs also have different spectra.

**Theorem 3.** *The cospectral regular pair of order 10 is not unique. Conjecture 00000002141 is false.*

*Remark 4.* The two pairs are not complements of each other: all four graphs are 4-regular, while complements of 4-regular graphs on 10 vertices are 5-regular. So uniqueness fails even up to complementation.

## 5 Enumeration

With nauty’s `geng`, `verify.py` enumerates all  $k$ -regular graphs on  $n \leq 10$  vertices and groups them by characteristic polynomial. There is no cospectral pair of regular graphs on fewer than 10 vertices. On 10 vertices there are exactly two cospectral pairs of 4-regular graphs, namely the two above (out of 60 4-regular graphs), and exactly two of 5-regular graphs, namely their complements. So the first clause (minimal order 10) is true, and there are exactly four cospectral regular pairs of order 10.

## 6 Lean formalization

The directory `lean4/` is a self-contained Lean 4.19.0 project (core library only, no Mathlib).

- Graphs on  $\{0, \dots, 9\}$  are Boolean adjacency functions. `IsSimple` means symmetric and loopless, and `Regular g k` means every vertex has degree  $k$ .
- `closedWalks g k` is  $\text{tr}(A^k)$ , computed with explicit matrix products. `Cospectral g h` means equal closed-walk counts for all  $k \leq 10$ , which by the argument above is equivalent to having the same adjacency spectrum.
- `OrthSimilar g h M` ( $MM^T = 4I$ ,  $MA_g = A_hM$ ) is proved for both pairs by `orth_1`, `orth_2`. `charCoeffs` applies Newton’s identities to the closed-walk counts, and `cp_A1`,  $\dots$ , `cp_B2` give the explicit coefficients.
- `Iso g h` is the existence of a bijection  $\sigma$  of  $\{0, \dots, 9\}$  (with inverse  $\tau$ ) such that  $g(u, v) = h(\sigma u, \sigma v)$ . `iso_hasNT` and `iso_hasIsolatedNT` prove that (P) and (Q) are isomorphism invariants. `invariants` evaluates (P) and (Q) on the four graphs.
- `CospectralRegularPair g h` combines simple, regular, cospectral and non-isomorphic. `UniquePair` says any two such pairs coincide up to isomorphism, in either order. `conjecture_00000002141_false` proves  $\neg \text{UniquePair}$ , using `pair_1`, `pair_2`, `not_iso_A1_A2` and `not_iso_A1_B2`. `conjecture_00000002141_conjunction_false` negates the whole conjunction for any first clause. As sanity checks, `iso_refl` and `iso_A1_A1r` (a nontrivial relabelling) show that `Iso` is not vacuous.

`lake build` succeeds. There is no `sorry`, no `native_decide`, and no added axioms. `#print axioms` shows at most `propext` and `Quot.sound`; the certificates use no axioms.

## 7 Reproduction

```
cd lean4 && lake build      # Lean 4.19.0, core only
python3 verify.py          # stdlib (+ nauty-geng if present)
pdflatex report.tex
```

## References

- [1] A. E. Brouwer and W. H. Haemers, *Spectra of Graphs*, Springer, 2012 (cospectrality and closed walks; GM switching).
- [2] W. H. Haemers and E. Spence, Enumeration of cospectral graphs, *European J. Combin.* 25 (2004), 199–211.